The wedge product of the coordinate differentials $\mathrm{d}q_i, \mathrm{d}p_i$ may be written

$$\mathrm{d}q_i \wedge \mathrm{d}p_i(\boldsymbol{\xi}, \boldsymbol{\eta}) = \xi_i \eta_{i+\mathrm{N_c}} - \xi_{i+\mathrm{N_c}} \eta_i = \boldsymbol{\xi}^T \boldsymbol{J}^{(i)} \boldsymbol{\eta},$$

where $\boldsymbol{J}^{(i)}$ is the matrix which has zeros everywhere except for $(\boldsymbol{J}^{(i)})_{i,i+\mathrm{N_c}} = 1$, $(\boldsymbol{J}^{(i)})_{i+\mathrm{N_c},i} = -1$. Summing these terms results in the *symplectic 2-form*, denoted ψ_{S}:

$$\psi_{\mathrm{S}} = \sum_{i=1}^{\mathrm{N_c}} \mathrm{d}q_i \wedge \mathrm{d}p_i\,(\boldsymbol{\xi}, \boldsymbol{\eta}) = \boldsymbol{\xi}^T \left(\sum_{i=1}^{\mathrm{N_c}} \boldsymbol{J}^{(i)} \right) \boldsymbol{\eta} = \boldsymbol{\xi}^T \boldsymbol{J} \boldsymbol{\eta}.$$

Thus the matrix

$$\boldsymbol{J} = \begin{bmatrix} \boldsymbol{0} & \boldsymbol{I} \\ -\boldsymbol{I} & \boldsymbol{0} \end{bmatrix}$$

defines a representation for the 2-form ψ_{S}. It is easy to verify that

$$\psi_{\mathrm{S}} = \sum_{i<j} \mathrm{d}z_i \wedge J_{ij} \mathrm{d}z_j.$$

In general, a differential 2-form ψ is represented in coordinates by

$$\psi_z = \sum_{i,j} a_{ij}(z) \mathrm{d}z_i \wedge \mathrm{d}z_j,$$

with matrix of coefficients $\boldsymbol{A}(z) = (a_{ij}(z))$.

The *pull-back* of a differential 1-form ψ under a phase space mapping $\boldsymbol{\Phi}$ is defined as the action of ψ after transformation by the Jacobian matrix of the mapping. It is written $\boldsymbol{\Phi}^* \psi$, so

$$(\boldsymbol{\Phi}^* \psi_z)(\boldsymbol{\xi}) = \psi_{\boldsymbol{\Phi}(z)}(\boldsymbol{\Phi}'(z) \boldsymbol{\xi}).$$

The pull-back of a differential 2-form $\psi_1 \wedge \psi_2$ is consequently defined as

$$\boldsymbol{\Phi}^*(\psi_1 \wedge \psi_2) = (\boldsymbol{\Phi}^* \psi_1) \wedge (\boldsymbol{\Phi}^* \psi_2).$$

Given a differential 2-form ψ_z represented by the matrix $\boldsymbol{A}(z) = (a_{ij}(z))$, the *pull-back* of ψ_z under $\boldsymbol{\Phi}$ is defined by

$$\boldsymbol{\Phi}^* \psi_z = \sum_{ij} b_{ij}(z) \mathrm{d}z_i \wedge \mathrm{d}z_j,$$